MILITARY SPECIFICATION SHEET

HEAT EXCHANGER, NBC SYSTEM
PRECOOLER P/N 12336768

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the precooler 12336768 described herein shall consist of this document and the latest issue of specification MIL-H-62411(AT).

3. REQUIREMENTS

3.4 Design and construction. The precooler shall conform to the drawing hierarchy of which Drawing 12336768 is the top drawing.

3.4.1 Weight. The weight of the precooler exchanger shall not exceed 42.9 pounds (lb) (see 4.6.3).

3.5 Performance.

3.5.1 Leakage.

3.5.1.1 External leakage. The pneumatic pressure applied to the hot air passages of the precooler shall be 100 pounds per square inch gage (psig) (see 4.6.4.1).

3.5.1.2 Internal leakage. The pressure applied to the hot air passages shall be 100 psig (see 4.6.4.2).

3.5.2 Proof pressure. The proof pressure shall be 200 psig (see 4.6.5).

3.5.3 Burst pressure. The burst pressure shall be 300 psig at 845 degrees Fahrenheit (°F) (see 4.6.6).

FSC 4130

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3.5.4 Heat transfer. Heat transfer performance shall be not less than that shown in figure 1 (see 4.6.7).

3.5.5 Air pressure loss. The pressure loss shall not exceed that shown in figure 2 (see 4.6.8).

4.6.4 Leakage.

4.6.4.1 External leakage. A pneumatic pressure of 100 to 103 psig shall be applied to the hot air passages.

4.6.4.2 Internal leakage. A penumatic pressure of 100 to 103 psig shall be applied to the hot air passages.

4.6.5 Proof pressure. The test pressure shall be 200 to 532 psig.

4.6.6 Burst pressure. The oven temperature shall be 845°F. The test pressure shall be 300 to 309 psig.

4.6.7 Heat transfer. To determine conformance to 3.5.4, the precooler shall operate at each combination of the steady state conditions specified in table VI. The measured hot air outlet temperature shall be equal to or greater than that specified in table VI.

TABLE VI. Heat transfer.

Initial air temperature		Air flow		Hot air Outlet temperature °F
Hot	Cold	Hot	Cold	
300 ± 15°F		10	15	173 ± 15
			30	276 ± 15
			52.2	290 ± 15
420 ± 20		16.6	15	271 ± 20
			30	336 ± 20
			52.2	374 ± 20
800 ± 40		20	15	460 ± 40
			30	600 ± 40
			52.2	680 ± 40

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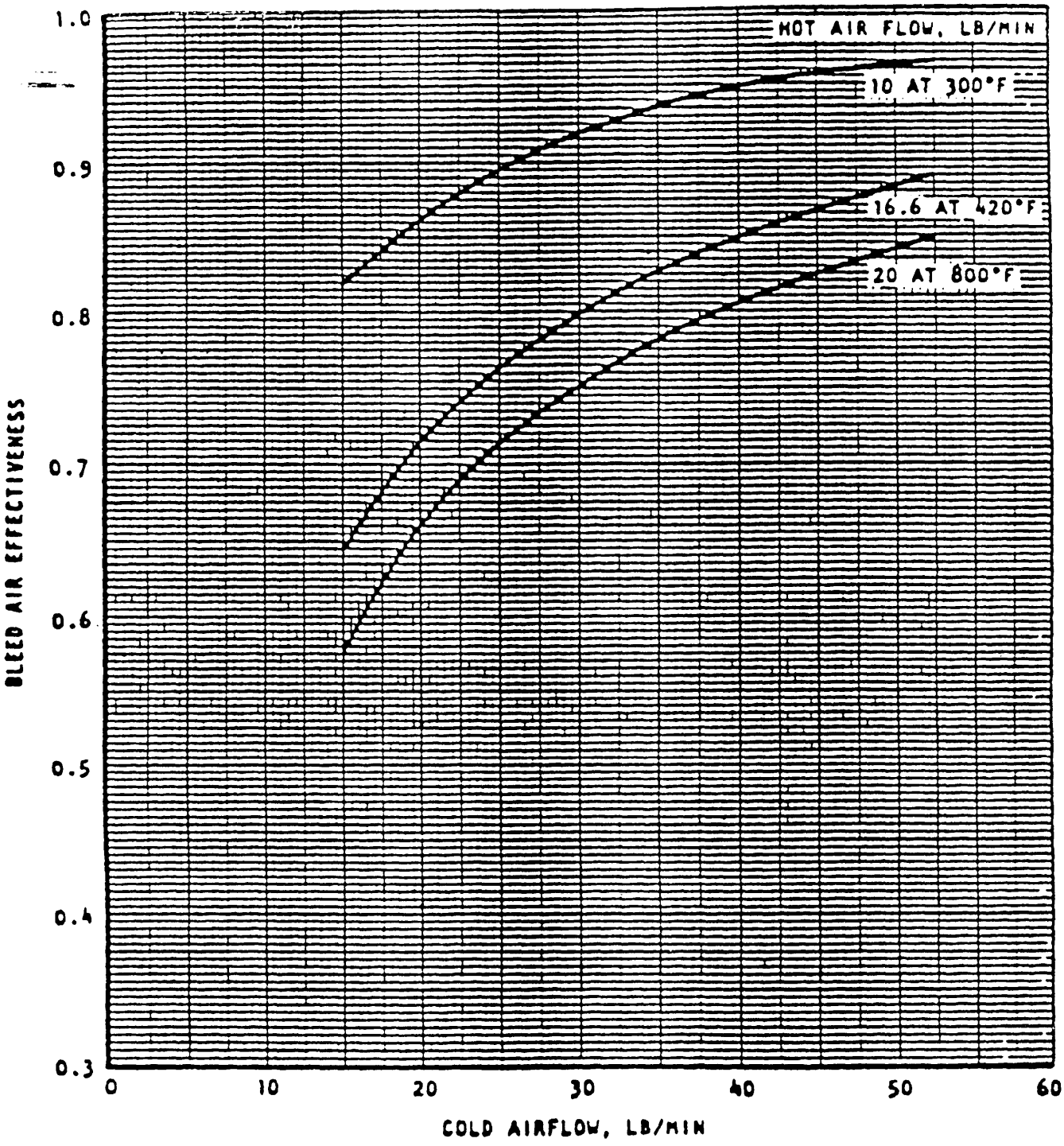


FIGURE 1. Required precooler performance.

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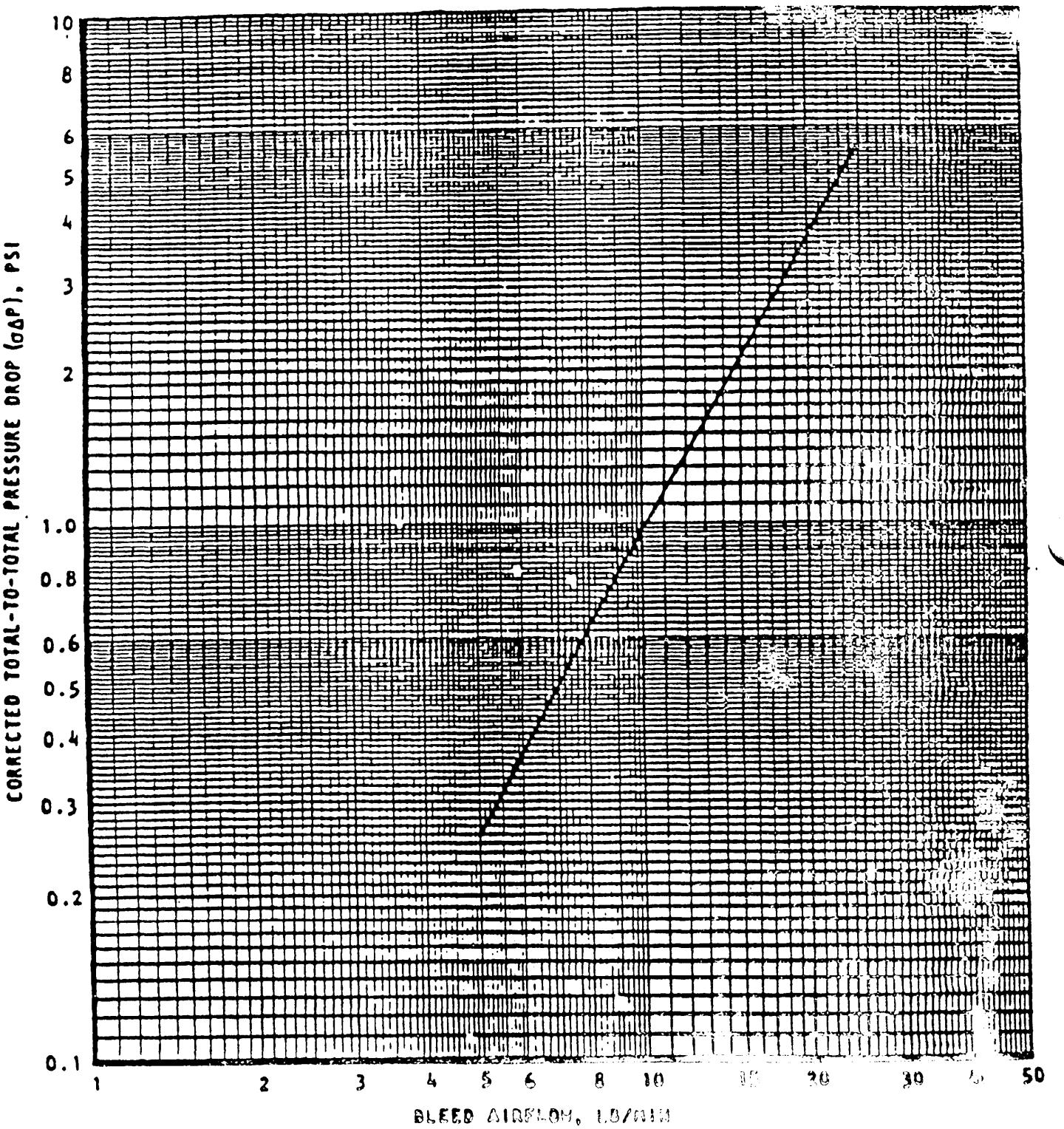


FIGURE 2. Allowable precooler bleed air pressure loss.

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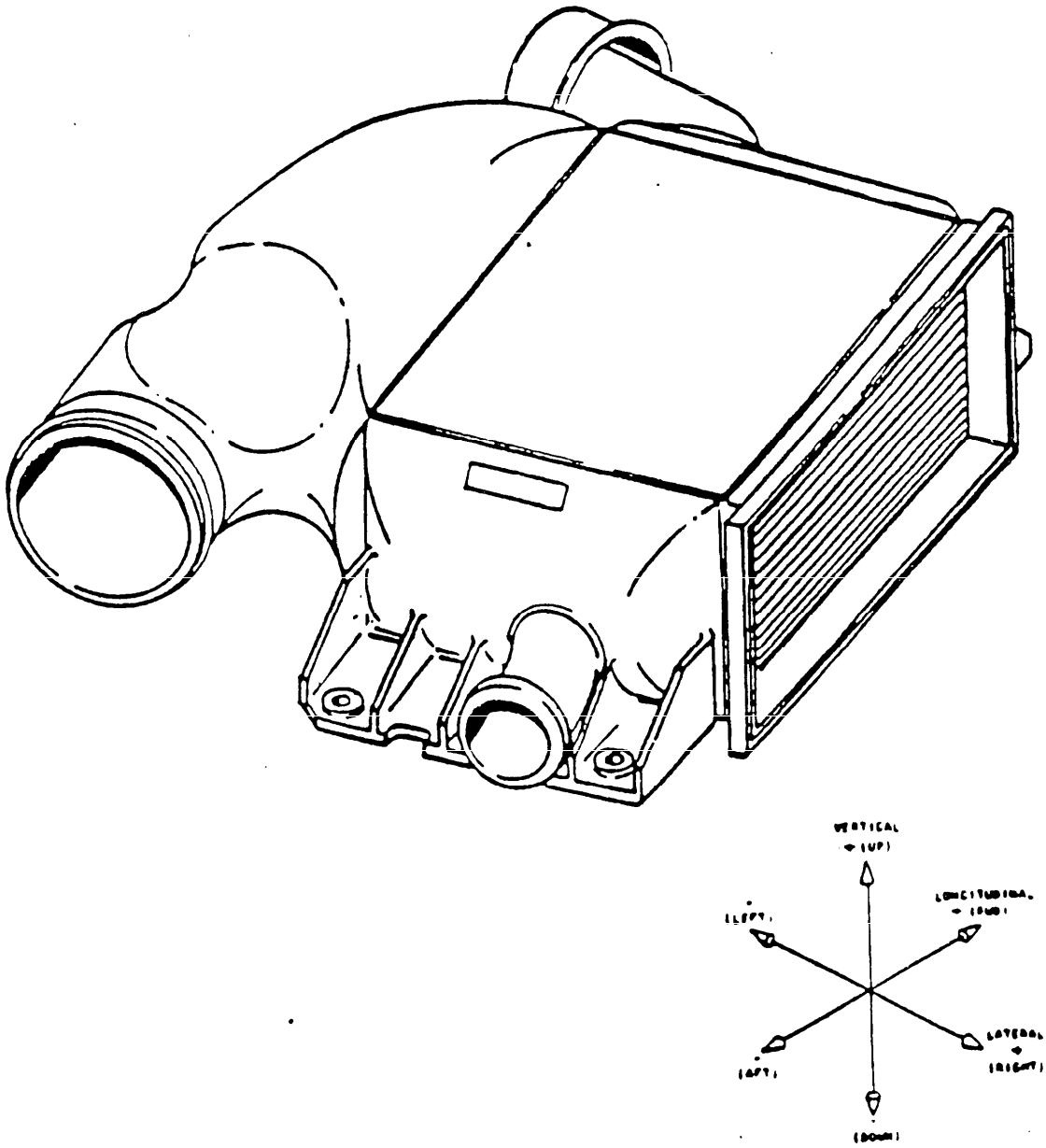


FIGURE 3. Axes orientation.

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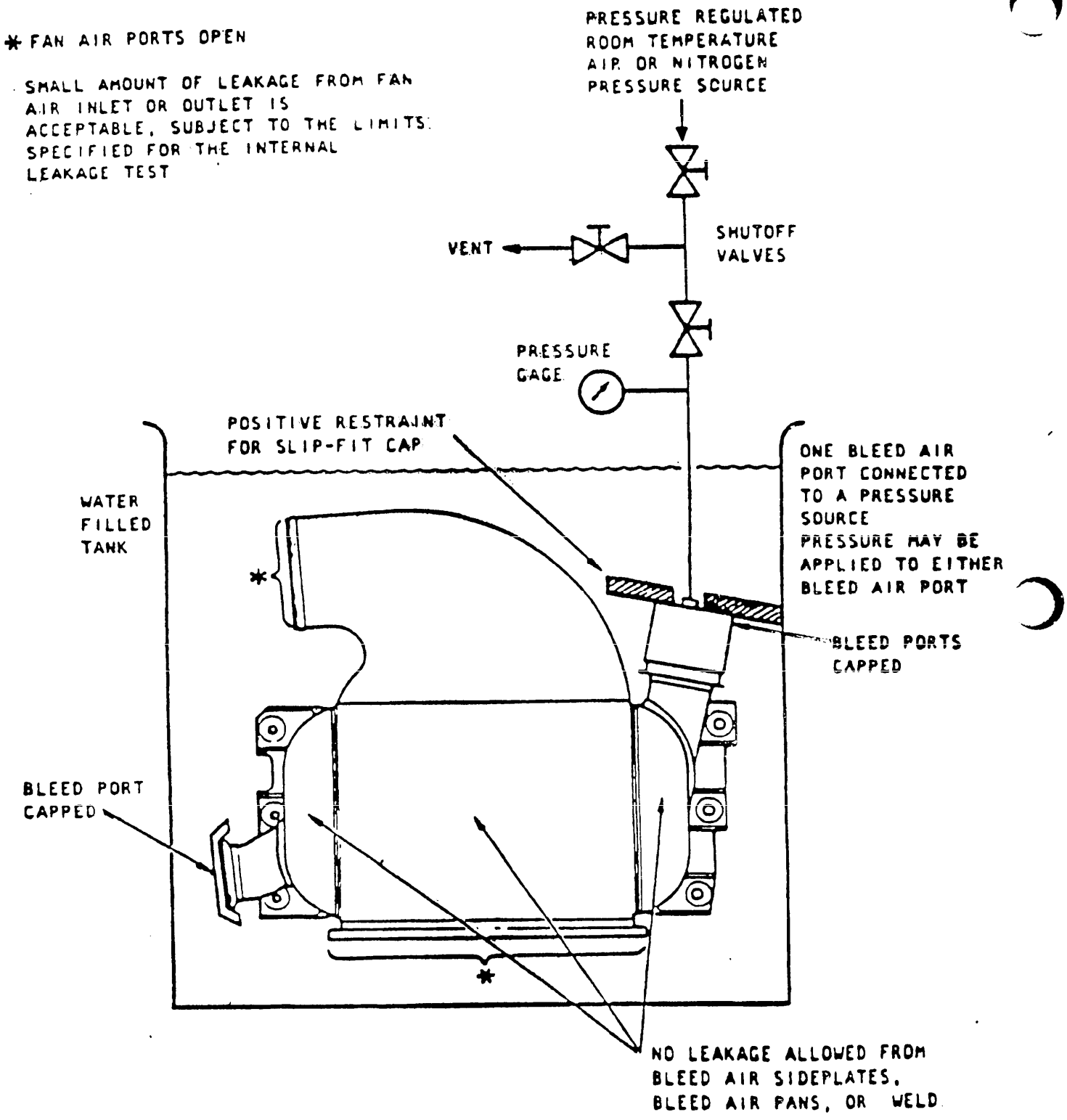


FIGURE 4. Leakage test setup.

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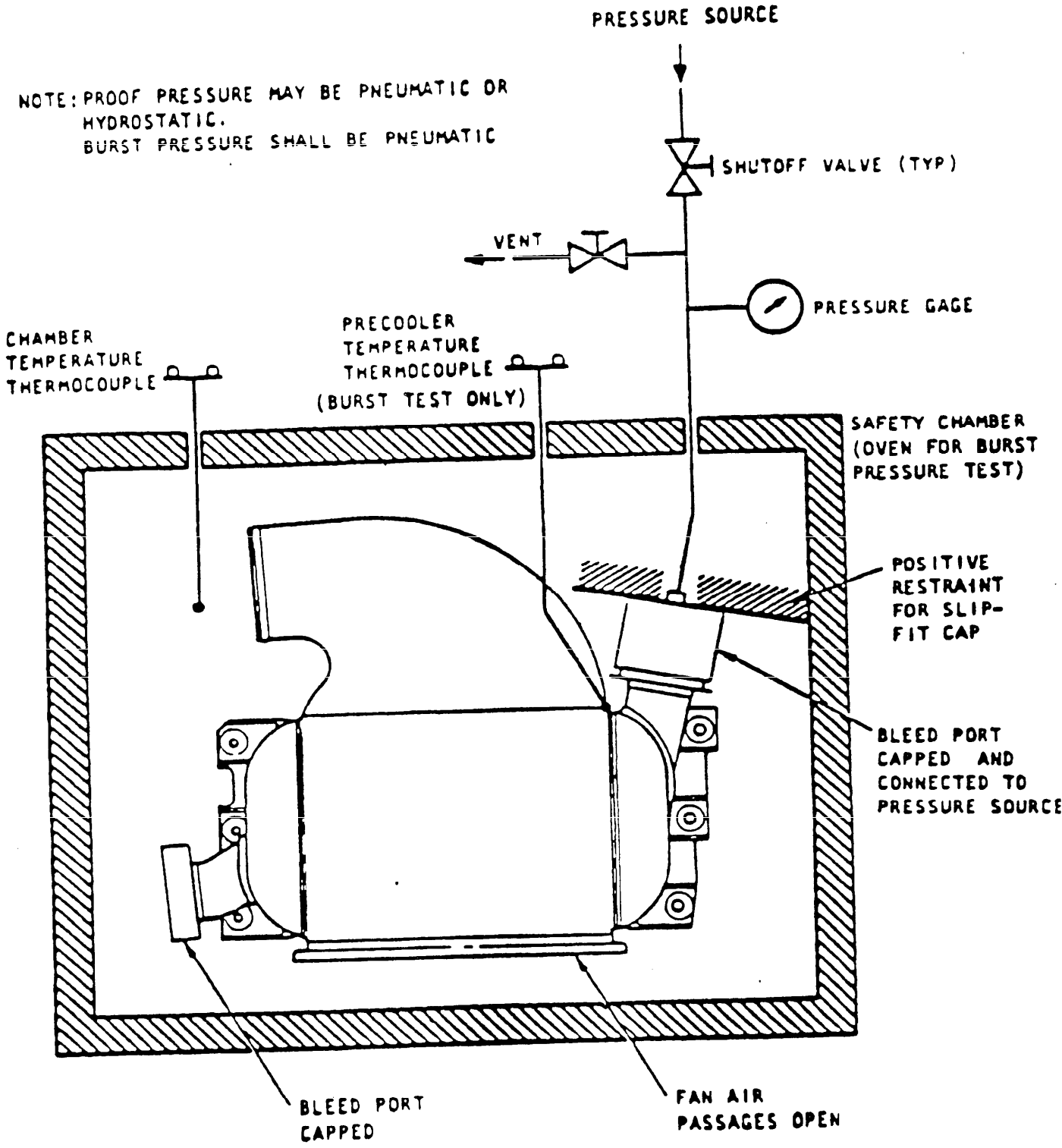


FIGURE 5. Proof and burst pressure test setup.

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